

3341.454 Mathematical and Numerical Optimization Final Exam

Spring 2026

June 15, 2026

3:30 - 4:45 pm

Question:	1	Total
Points:	100	100

1. Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ and $g : \mathbb{R}^m \rightarrow \mathbb{R}$ be closed convex functions. Let $A \in \mathbb{R}^{m \times d}$ be a matrix. We consider the following composite convex minimization problem:

$$\text{minimize} \quad f(x) + g(Ax). \quad (\text{P})$$

- (a) (20 points) Let f^* and g^* be the Fenchel conjugates of f and g , respectively. Derive the dual of (P).
- (b) (40 points) Deduce the dual subgradient method by applying the subgradient method to the dual of (P).
- (c) (40 points) Show that the proximal point algorithm applied to the dual of (P) is equivalent to the augmented Lagrangian method.